

LIGTAS User Manual

Updated 3 Oct 2026

LIGTAS tells each resident what to do for their own purok when a flood alert is signed, including when the internet is down; this manual covers the resident app and how to run the demo.

Source code: github.com/Raizellrou/ligtas · **Live demo:** pwa-five-mocha.vercel.app

Getting started

Open the app, tap Continue on the welcome screen, and pick your purok; the app remembers it.

1. Open the app. On a barangay hub, connect to the hub's WiFi and use the page it serves. The hosted demo is at pwa-five-mocha.vercel.app and shows a recorded run.
2. Tap **Continue** on the welcome screen.
3. Choose your purok from the grid of 1 to 12, then tap **Select a purok to continue**. You can change it any time.
4. The **Resident** tab now shows the alert for your purok. With no alert it says "Your purok is not affected by any current alert."

The app has up to three tabs: **Resident** is the real screen, **How it works** is a short explainer, and **Tester** appears when you tap "Open the simulator" at the bottom of the purok screen (see The Tester tab).

Reading an alert

Every alert has a severity tier from 1 to 3, and the tier decides how loud the screen is.

Tier	Level	What you see
1	Watch	A quiet note on the home screen
2	Prepare	A marigold card on the home screen
3	Evacuate	A full-screen takeover naming the nearest center and the way there

Only Tier 3 interrupts you. It cannot be closed with the Escape key; tap **Show my route** to continue, and the same alert will not come back. The newest alert for your purok wins, so a later lower tier means the danger is easing.

Tier 3 vibrates the phone once and makes no sound. Chrome only allows vibration after you have tapped the page, and iPhone browsers do not support it, so do not rely on it; the siren is the primary warning.

Each alert shows when it was issued, for example "Issued 25 min ago · 3:42 PM". The app checks the signature on every alert itself before showing it.

Evacuation map and route

The map shows Nangka, Marikina and the walking route from your purok to the nearest evacuation center, and it draws with no connection.

- **Your position (optional).** Tap to show where you are. Your browser asks permission only then; the location is used on your phone to reroute and is never sent anywhere. It needs HTTPS or `localhost`.
- **Flood-aware routes.** From Tier 2 up, the route avoids streets close to rivers, streams and canals. If every route looks flooded, the app says so, still shows the ordinary route as a warning, and tells you to follow barangay officials.
- **Read the labels.** The map says "Demo layout, not official". Purok positions are a generated layout, the centers are places OpenStreetMap names and are not confirmed by the LGU, and the flood model is a stand-in, not a survey. Map data is © OpenStreetMap contributors (ODbL).

Household check-in

Joining a household is optional: alerts and the map only need a purok, so you can skip it.

1. On the Resident tab, choose to join your household.
2. Enter the household join code your barangay gave you. For the demo, codes are in `packages/hub/config/households.json`.
3. Type your name and tap **Continue**.

Once joined you can tap **I'm safe** or **I need help**, and see your family's statuses. A tap is saved on your phone first and sent to the hub when it can be, so it is safe to tap while offline. Each member shows how long ago the hub recorded their status; that is when the hub heard it, not when the person tapped.

While your household has a live alert against it, a relief pill shows its claimable balance, read straight from Stellar Testnet.

Joining needs a hub. On the hosted demo there is none, so joining shows "Can't reach the hub right now."

Offline and stale information

An offline phone can show hours-old information, so the app says how old each thing is.

- **Alerts** show when they were issued, and after 12 hours add "this alert is old".
- **The alert list** shows when your phone last got an update, for example "Alerts checked 4 min ago". After 3 hours, or 10 minutes when connected to a live hub, it becomes a warning that you may have missed newer alerts, so "not affected" is never shown bare from a phone that is out of touch.
- **After one visit**, the app, the map and the last alerts your phone received are stored, so it still opens with no network. When the connection returns, the offline banner clears.
- **There is no push**. With a live hub the app checks every 15 seconds, but only while it is open and visible. A phone in a pocket hears nothing and depends on the siren.

Running the demo

To see live alerts, run the hub and the app on one computer; you need Node 20 or newer and pnpm.

1. Install and build once (the hub uses the built `core` and `stellar` packages):

```
pnpm install
pnpm --filter @ligtas/core build
pnpm --filter @ligtas/stellar build
pnpm --filter @ligtas/hub build
```

2. Start the hub. It listens on port 3001:

```
node packages/hub/dist/index.js
```

3. In a second terminal, start the app:

```
pnpm --filter @ligtas/pwa dev
```

4. Open `http://localhost:5173`. It must be exactly that address, because the hub only answers that origin (`LIGTAS_PWA_ORIGIN`); the production preview on port 4173 will not reach it.
5. Send a signed alert to the hub's `/alert` route using `packages/core/scripts/emit-alert.ts` (see `packages/hub/README.md`). The open app shows it within about 15 seconds.

To also anchor alerts and release payouts on Stellar Testnet, start the hub with `LIGTAS_HUB_STELLAR_SECRET` set to a funded Testnet secret. Without it the hub still accepts and serves alerts but does not anchor or pay. Never commit that secret.

Before presenting, rebuild the hub on the machine you will present from. The hub keeps its history in `hub.sqlite`, which is ignored by git and carries over between runs.

The Tester tab

The Tester stands in for the sensor hardware and shows how the app treats good and bad packets. Open it from the link at the bottom of the purok screen. Every packet is really built and signed in your browser, and the Resident view decides what each one is by checking it.

Button	What it sends	Expected result
Genuine alert	Signed by this browser's authorised issuer key	Accepted and shown
Forged alert	Really signed, but by an impostor keypair the verifier does not trust	Rejected
Tampered alert	Signed correctly, then one body byte flipped	Rejected
Replay (old sequence)	A fresh, correctly signed packet reusing sequence 1	Rejected

Other controls:

- **River gauge.** The slider runs from 0 to 400 cm; Tier 1 starts at 150 cm, Tier 2 at 200 cm and Tier 3 at 250 cm.
- **Transmission log.** Lists what was sent from here. "Show full verification log" shows every packet, including the captured demo run.
- **Reset.** Drops everything broadcast here and returns to the captured run.
- **Walk simulator.** A fake position walks from your purok to the nearest center; raise the river mid-walk to watch the route change.
- **Live mesh demo.** Runs the real simulated radio mesh, but only on a hub started with `LIGTAS_ENABLE_MESH_ORCHESTRATION=1` and a Meshtasticator checkout. Otherwise the panel lists what is missing.

Troubleshooting and known limits

What you see	Likely cause	What to do
"Can't reach the hub right now"	The hub is not running, or the app is not on the address the hub allows	Start the hub and open <code>http://localhost:5173</code> . On the hosted demo there is no hub, so this is expected when joining a household.
"Demo data, not a live hub" banner	The app is showing the recorded run from 12 Sep 2026	Expected on the hosted demo. Run the hub locally for live alerts.
"Your purok is not affected"	No accepted alert includes your purok	Check the "Alerts checked" time. If it is old, reconnect before trusting it.
Warning that you may have missed newer alerts	No contact with the hub for 3 hours, or 10 minutes on a live hub	Reconnect, and follow the siren and barangay officials.
Location does not appear	Permission not granted, or the page is not on HTTPS or localhost	Allow location when asked.
No vibration on a Tier 3 alert	Chrome needs a tap first, and iPhone browsers do not support it	Rely on the siren and the on-screen takeover.
Live mesh demo says it is unavailable	Orchestration is off, or Meshtasticator is missing	Optional. Set <code>LIGTAS_ENABLE_MESH_ORCHESTRATION=1</code> and install Meshtasticator with Docker.

Known limits, stated plainly:

- The radio mesh is simulated, so real-terrain range is not yet measured.
- The map is a demo snapshot: purok positions and the flood model are not a survey.
- Payouts are parametric: by location and threshold, not by verified damage.
- Everything runs on Stellar Testnet only; no real money moves.